

Microsoft Certified: Security Operations Analyst Associate

Course

Defend against cyberthreats with Microsoft's security operations platform

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Module 2

Mitigate threats using Microsoft Security Copilot

Modules in this learning path

COURSE 2

Describe Microsoft Security Copilot

Introduction

Organizations face unprecedented challenges, in the rapidly evolving digital landscape, such as increasingly sophisticated cyber threats and a significant talent shortage in cybersecurity. Microsoft Security Copilot is a cloud-based, AI-powered security analysis tool that is designed to address these challenges. It enables analysts to process security signals and respond to threats at a machine speed that far surpasses human capabilities, thus revolutionizing the way organizations approach cybersecurity.

After completing this module, you'll be able to:

Describe what Microsoft Security Copilot is.

Describe the terminology of Microsoft Security Copilot.

Describe how Microsoft Security Copilot processes prompt requests.

Describe the elements of an effective prompt.

Describe how to enable Microsoft Security Copilot.

Get acquainted with Microsoft Security Copilot

The top security challenges organizations face include:

An increase in the number and sophistication of attacks.

A talent shortage that's driving the need for automation, integration, and consolidation of security tools.

Visibility into security, privacy, compliance, and governance.

Organizations need to act quickly to address the security challenges they face, but working at human speed, even if there weren't a talent shortage, isn't enough. Organizations need to work at machine speed.

Microsoft Security Copilot is an AI-powered, cloud-based security analysis tool that enables analysts to respond to threats quickly, process signals at machine speed, and assess risk exposure more quickly than may otherwise be possible.

Use cases

Security Copilot focuses on making the following highlighted use cases easy to use.

Investigate and remediate security threats - gain context for incidents to quickly triage complex security alerts into actionable summaries and remediate quicker with step-by-step response guidance

Build Kusto Query Language (KQL) queries or analyze suspicious scripts - eliminate the need to manually write query-language scripts or reverse engineer malware scripts with natural language translation to enable every team member to execute technical tasks

Understand risks and manage security posture of the organization - get a broad picture of your environment with prioritized risks to uncover opportunities to improve posture more easily

Troubleshoot IT issues faster - synthesize relevant information rapidly and receive actionable insights to identify and resolve IT issues quickly

Define and manage security policies - define a new policy, cross-reference it with others for conflicts, and summarize existing policies to manage complex organizational context quickly and easily

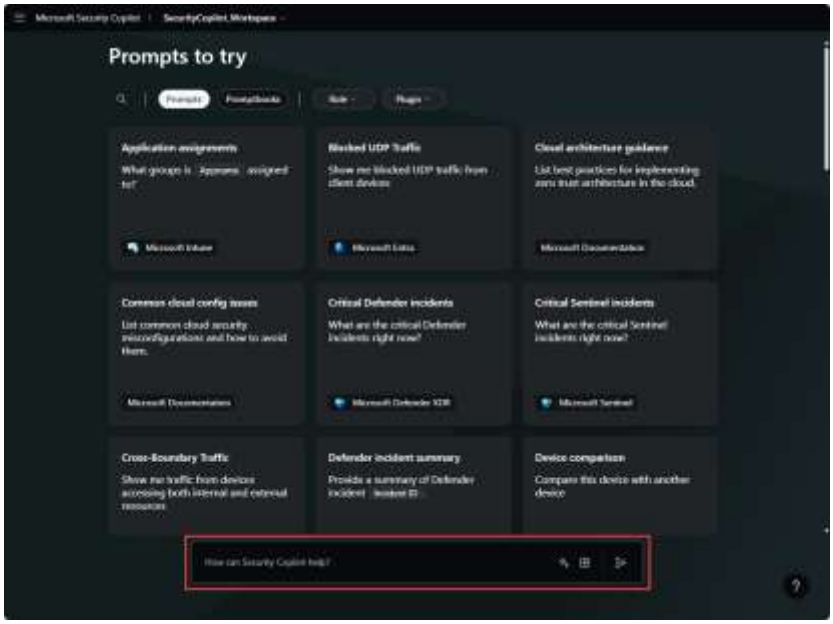
Configure secure lifecycle workflows - build groups and set access parameters with step-by-step guidance to ensure a seamless configuration to prevent security vulnerabilities

Develop reports for stakeholders - get a clear and concise report that summarizes the context and environment, open issues, and protective measures prepared for the tone and language of the report's audience

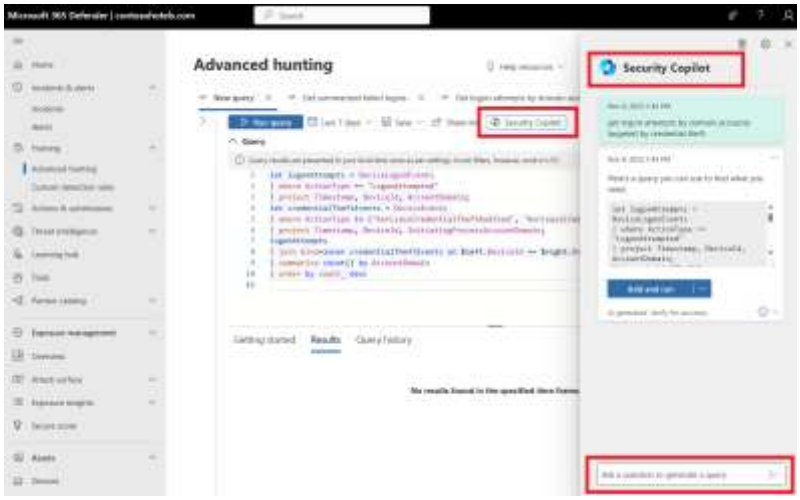
These use cases represent just a few of the capabilities that Copilot delivers and that helps make analysts more productive and also helps up-level them.

Standalone and embedded experience

You can experience Copilot through the dedicated site, also referred to as the standalone experience. Users interact with Copilot through the prompt bar. In the prompt bar, users make requests in natural language and receive response outputs as text, images, or documents.



Additionally, some Microsoft security products embed Copilot capabilities directly within the products' user interface. This experience is referred to as the embedded experience. Microsoft Defender XDR, for example, enables Copilot capabilities including summarizing incidents, analyzing scripts, generating KQL queries, and more.



More information on both the standalone and embedded experience are covered in subsequent modules. Images shown throughout the rest of this module are based on the standalone experience.

Watch this short video for a summary of the users' experiences that Security Copilot offers.

Natural language processing (NLP)

Copilot is built using Azure OpenAI Services and is designed to integrate with existing security tools and processes, making it easier for organizations to improve their overall security posture. Azure OpenAI Services provides REST API access to OpenAI's powerful large language models (LLMs) for natural language processing (NLP), while providing security capabilities of Microsoft Azure.

With access to the powerful LLMs for NLP, Copilot is able to read, decipher, and make sense of human languages, enabling users to securely interact with it using natural language. Although the LLMs are trained on a vast amount of information that endows Copilot with broad general knowledge and problem solving abilities, it’s not enough. Security analysts expect their copilot to be trained on security and that's where the integration with existing security tools and processes comes into play.

Integration with Security-specific sources

Copilot combines powerful LLMs with security-specific sources from Microsoft. These security-specific sources are informed by Microsoft’s unique global threat intelligence, more than 65 trillion daily signals, and incorporates information from a growing set of security solutions using plug-ins and connections to knowledge bases. Through plug-ins, Copilot integrates with Microsoft's own security products, non-Microsoft products, and open-source intelligence feeds. Connections to an organization's knowledge bases gives Copilot more context, resulting in responses that are more relevant, specific, and customized to the user. Through the powerful combination of advanced general models and security specific sources, Copilot is able to learn at machine speed to help analysts identify and respond to emerging threats.

The information you give Copilot is only accessible to your organization. Your data is your data, and it's protected by comprehensive enterprise compliance and security controls. Your data isn't used to train the foundation AI models.



Microsoft Security Copilot is the first security product to enable defenders to move at the speed and scale of AI.

Describe Microsoft Security Copilot terminology

In this unit, we introduce you to some basic terminology.

Terminology

The following terms are important for understanding the way Microsoft Security Copilot works:

Session – A particular conversation within Copilot. Copilot maintains context within a session.

Prompt – A specific statement or question within a session. A user enters a prompt in the prompt bar.

Capability – A function Copilot uses to solve part of a problem. A capability may sometimes be referred to as a skill.

Plugin – A collection of capabilities by a particular resource.

Workspace - Copilot workspaces are separate Copilot work environments within the tenant in which your Copilot instance is operating.

Agents - Microsoft Security Copilot agents are AI-powered tools that autonomously manage security and IT tasks, enhancing threat response, reducing manual workloads, and improving efficiency across cybersecurity operations at scale.

Orchestrator – Copilot's system for composing capabilities together to answer a user's prompt.

Prompt bar and sessions

At the center of Security Copilot is the prompt bar. You use the prompt bar to tell Copilot what insights you want from your security data, this is referred to as the prompt. In other words, the prompt is the text-based, natural language input you provide in the prompt bar that instructs Copilot to generate a response. Although you interact with Copilot in natural language, it's helpful to be specific in the prompts (specific questions or statements) that you provide. For those that are relatively new to the security analyst role and engaging with AI, effective prompting may take some practice. For this reason, Copilot provides promptbooks that provide a series of preselected prompts and prompt suggestions (more details on this in a subsequent module).



As you make requests and as Copilot responds, you might have some follow-up requests. The entirety of the dialog is referred to as a session. Copilot maintains context within a session.

Plugins and capabilities

In the previous unit, we mentioned that Copilot integrates with various sources through plugins, including Microsoft's own security products such as Microsoft Sentinel, Microsoft Defender XDR, and Microsoft Intune, non-Microsoft solutions, and open-source intelligence feeds. The integration enabled by the plugin, for any specific data source, provides Copilot with a collection of capabilities. Each capability is like a function in software, it's designed to do a specialized task within the scope of the data source. For example, the plugin to Microsoft Defender XDR includes a collection of individual capabilities that are used only by Microsoft Defender XDR. These include:

The ability to summarize an incident.

Support incident response teams in resolving incidents through guided responses (a set of recommended actions based on the specific incident).

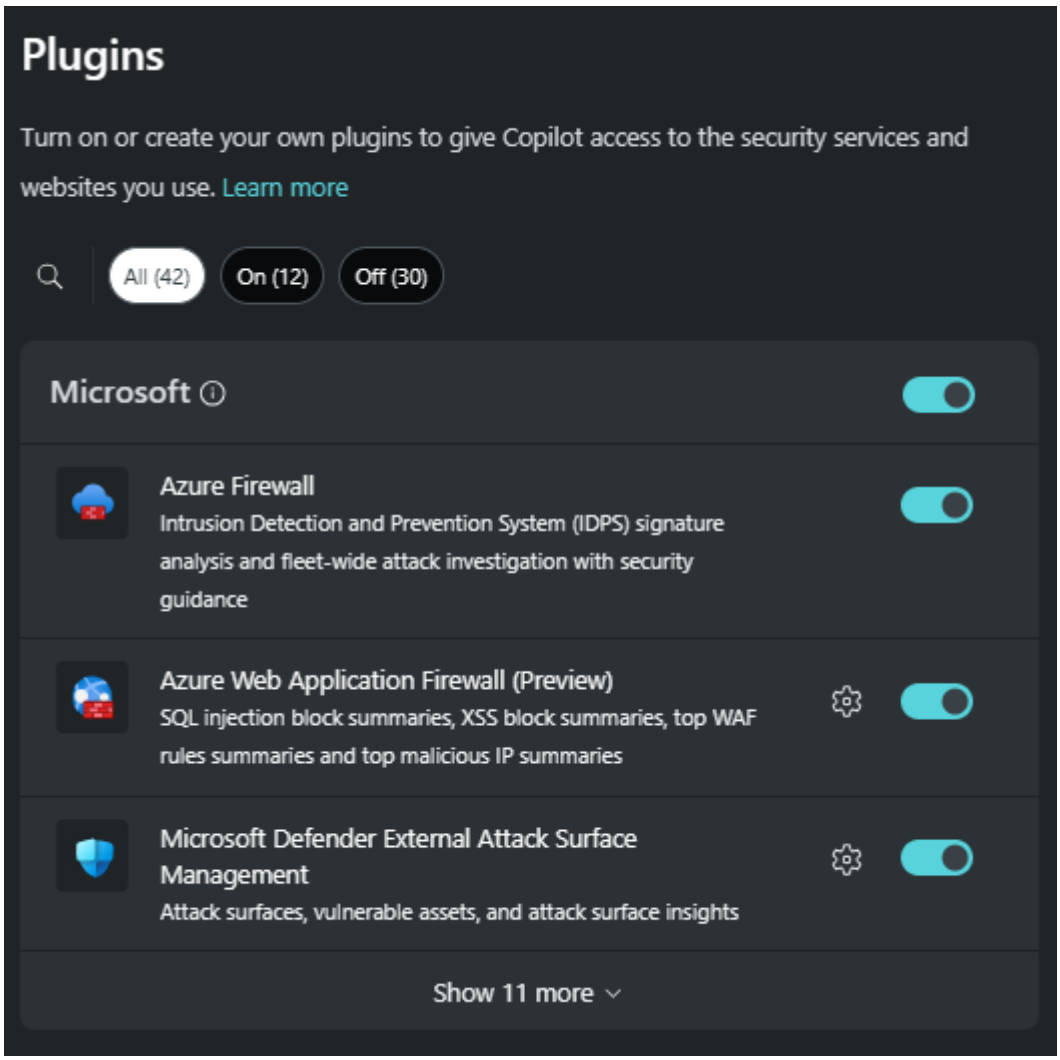
The ability to analyze scripts and code.

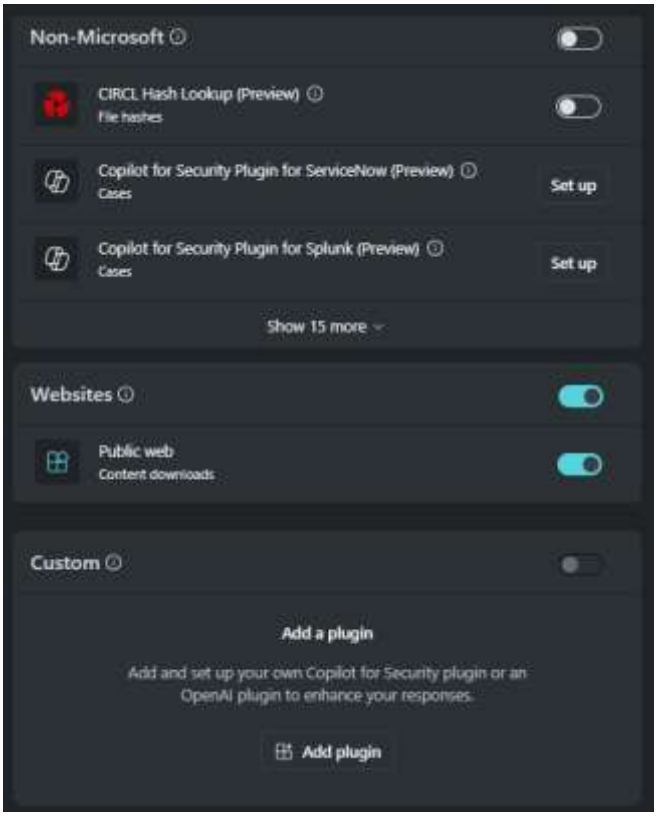
The ability to generate KQL queries from natural language input.

The ability to generate incident reports.

A plugin for Microsoft Sentinel might have similar capabilities but runs only within the scope of Microsoft Sentinel.

Copilot currently supports plug-ins for Microsoft services and non-Microsoft services, including websites and custom plug-ins that can be enabled.



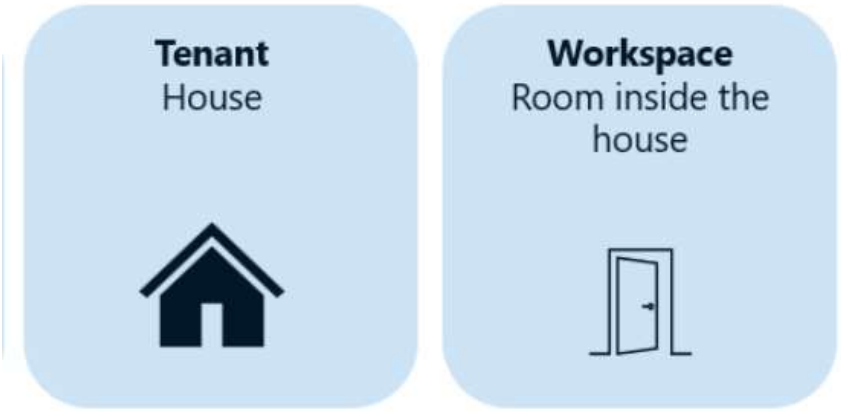


Some plugins require setup and configuration, as depicted by the Setup button or the gear icon. For Microsoft plugins, set up might be required where resource specific information needs to be specified. For non-Microsoft sources, set up might be required for account authentication.

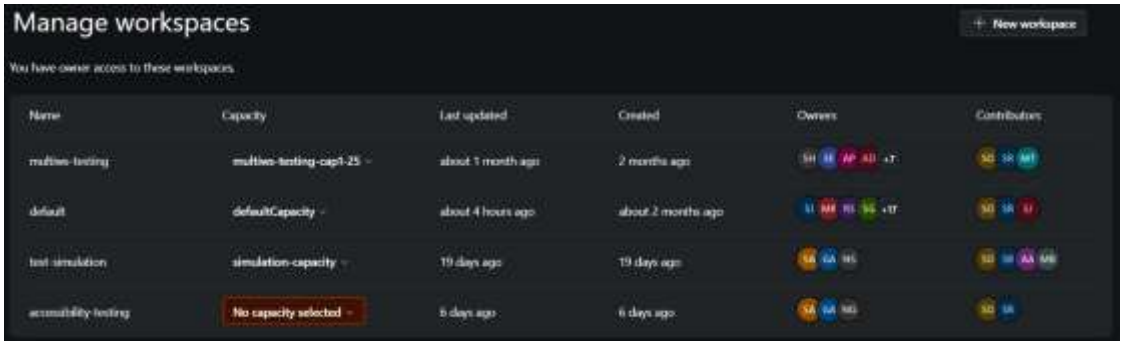
Workspaces

Copilot workspaces are separate Copilot work environments within the tenant in which your Copilot instance is operating.

To help you better understand the concept of workspaces, we use the analogy of house with multiple rooms. Each room is configured to be optimized for its function and the people that use that room. When someone enters the house, they might have access to some rooms but not others.



You can think of Copilot Workspaces fitting into this analogy. A Copilot workspace is analogous to a room in a house. You can also think of the house as analogous to a tenant. In the same way that a house has multiple rooms, the tenant in which Copilot is operating can have multiple workspaces.



Through the tenant-switching capability in Security Copilot, a user can select in which tenant they'll be working. In our analogy, this is a Copilot user getting access to the house. Once the tenant is selected, a Copilot user can access and work in any workspace (room in the house) to which they have access, within the context of their role permissions in that workspace.

Workspaces are powered by capacities and each workspace must have its own capacity.

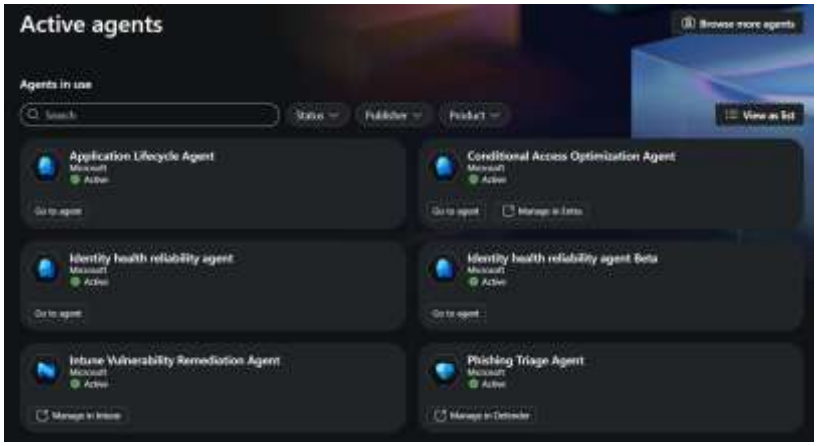
Using workspaces, you can efficiently map and monitor costs based on team needs and budgets, ensuring that teams have the capacity they need and resources are allocated effectively. Having workspaces also allows you to store session data according to geo-specific regulations and adhere to local data protection laws. These are just a few of the benefits of using workspaces.

For more information, see "Describe workspaces", which is linked in the Summary and resource section of this module.

Agents

A Microsoft Security Copilot agent is an advanced, AI-powered assistant built into Microsoft Security Copilot. These agents go beyond just answering questions—they can autonomously manage high-volume security and IT tasks. They're deeply integrated with Microsoft's security tools and can also work with partner solutions. Each agent is tailored for specific security scenarios, such as threat protection, identity management, or data security.

These agents are designed to learn from feedback, adapt to your organization's workflows, and operate securely within Microsoft's Zero Trust framework. See the summary and resources unit for links to more information on Microsoft Security Copilot agents.



Orchestrator

The orchestrator is Copilot's system for composing capabilities together to answer a user's prompt. This function is illustrated in more detail in the subsequent unit that describes how Copilot processes prompt requests.

Describe how Microsoft Security Copilot processes prompt requests

So now that there's a basic understanding of plugins, capabilities, and how the user interacts with Microsoft Security Copilot through prompts, it's worth taking a look under the hood to see how these components come together to process a prompt request and help security analysts.

Process flow

When a user submits a prompt, Copilot processes that prompt to generate the best possible response. The diagram that follows illustrates, at a high level, steps that Copilot takes to process the prompt and generate a response.



Submit a prompt: The process starts when a user submits a prompt in the prompt bar.

Orchestrator: Security Copilot sends the information to the Copilot backend referred to as the orchestrator. The orchestrator is Copilot's system for composing capabilities together to answer a user's prompt. It determines the initial context and builds a plan using all the available capabilities (skills).

Build context: Once a plan is defined and built, Copilot executes that plan to get the required data context to answer the prompt.

Plugins: In the course of executing the plan, Copilot analyzes all data and patterns to provide intelligent insights. This includes reasoning over all the plugins and sources of data, enabled and available to Copilot.

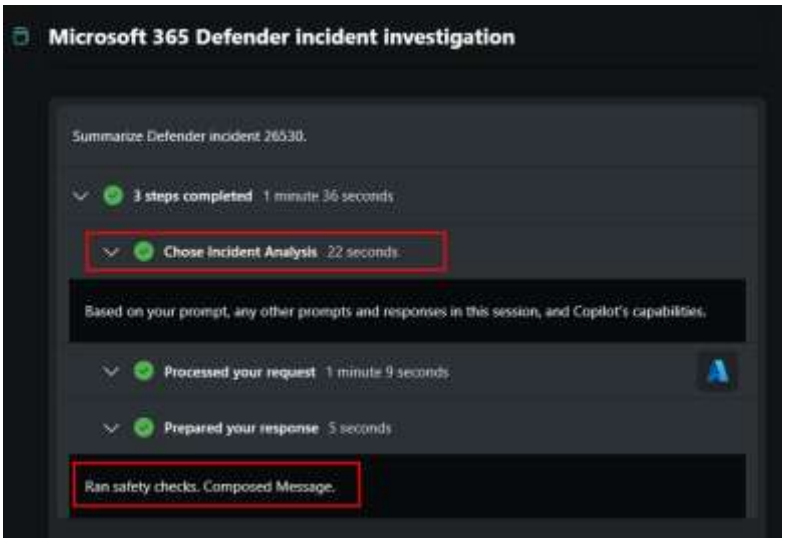
Responding: Copilot combines all the data and context and uses the power of its advanced LLM to compose a response using language that makes sense to a human being.

Response: Before the response can be sent back to the user, Copilot formats and reviews the response as part of Microsoft's commitment to responsible AI.

Receives response: The process culminates with the user receiving the response from the Copilot.

Process log

During this process, Copilot generates a process log that's visible to the user. The user can see what capability is used to generate the response. This is important because it enables the user to determine whether the response was generated from a trusted source. In the screenshot that follows, the process log shows that Copilot chose the Incident Analysis capability. The process log also shows that the final output went through safety checks, which is part of Microsoft's commitment to responsible AI.



Describe the elements of an effective prompt

In a previous unit, we defined a prompt as the text-based, natural language input you provide in the prompt bar that instructs Microsoft Security Copilot to generate a response. Copilot provides promptbooks and prompt suggestions, which are helpful, particularly if you're just starting an incident investigation. At some point, however, you'll want and need to enter your own prompts. In those cases, the quality of the response that Copilot returns depends in large part on the quality of the prompt used. In general, a well-crafted prompt with clear and specific inputs leads to more useful responses by Copilot.

Elements of an effective prompt

Effective prompts give Copilot adequate and useful parameters to generate a valuable response. Security analysts or researchers should include the following elements when writing a prompt.

Goal - specific, security-related information that you need

Context - why you need this information or how you use it

Expectations - format or target audience you want the response tailored to

Source - known information, data sources, or plugins Copilot should use



Every good prompt should have a goal. Whether it comes in the form of instructions or questions, it should indicate what you want out of your current session.

For Copilot, context can refer to the time frame, or that you use the response for a report. Expectations can include whether you want the response to be in a table format, a list of action steps, a summary, or even a diagram. Source might be useful in specifying which Microsoft plugins you're referring to, if needed. Some plugins require more context to work effectively or supporting plugins to ensure a response when initial responses fail.

Other prompting tips

Some things to remember when coming up with your own prompts:

Be specific, clear, and concise as much as you can about what you want to achieve. You can always start simply with your first prompt, but as you get more familiar with Copilot, include more details following the elements of an effective prompt.

Basic prompt: Pearl Sleet actor

Better prompt: Can you give me information about Pearl Sleet activity, including a list of known indicators of compromise and tools, tactics, and procedures (TTPs)?

Iterate. Subsequent prompts are typically needed to further clarify what you need or to try other versions of a prompt to get closer to what you're looking for. Like all LLM-based systems, Copilot can respond to the same prompt in slightly different ways.

Provide necessary context to narrow down where Copilot looks for data.

Basic prompt: Summarize incident 15134.

Better prompt: Summarize incident 15134 in Microsoft Defender XDR into a paragraph that I can submit to my manager and create a list of entities involved.

Give positive instructions instead of "what not to do." Copilot is geared toward action, so telling it what you want it to do for exceptions is more productive.

Basic prompt: Give me a list of unmanaged devices in my network.

Better prompt: Give me a list of high-risk unmanaged devices in my network. If they're named "test" remove them from the list.

Directly address Copilot as "You" as in, "You should ..." or "You must ...," as this is more effective than referring to it as a model or assistant.

While these guidelines can help you get started in creating prompts, it’s important to note that you’re not limited to forming prompts following the structure of the previous examples. What’s great about Copilot is that it's designed to respond to questions or instructions made in your own words (that is, using natural language).

You have the flexibility to adapt these guidelines to your specific needs.